

BLAST FURNACE PRODUCTION SECTION													
DATE-	1. Jan. 2025	DAILY PARAMETERS REPORT										ON-DT	ON-Month
		ABP	OPP			EB	900			Crude steel To HM Ratio		0.960	0.960
Monthly Plan (T)		599000	550000	Sinter Prod. / Nut Coke		8120	50	STOCKS		ON-DT		Prev.Day	
Daily Rate (T/Day)		17692		SP - II		8820		Sinter stock		High lines (1-7)		2000	1800
H.M.SENT TO SMS's		17443		SP - III		9270		I/Ore lump stock		Em.Storage		17400	18300
				COAL STOCK		191419	M/c-II			OHP		8705	8190
				Dry Coke ==>		7748	TOTAL	26210		OHP # 2		29609	26030
										BF		60974	61162
										CO		24023	24123
										Coke stock supply from stock		100	200
										BFs			
										FINE COAL		880	812
										RAW COAL		360	250
										CDI STOCK (1 - 7)			
										SHIPPING REPORT			
										HM (PCM)		249	230.65
										PROD.		230.65	0.00
										BY ROAD			0.00
										BY RAIL			
										Total Desp.		6701.90	6701.90
										STOCK		3012921.46	
										No.OF TRIPS		161	

HOT METAL AND SLAG QUALITY

Parameter	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8	SHOP	TILL	(BF-4 to 7)
Avg. % 'Si'		0.60	0.92	0.78	0.67	0.71	0.71	0.71	0.73
Min/Max		0.17/0.83	0.70/1.13	0.21/1.02	0.22/1.25	0.43/0.99			
Till % 'Si'		0.60	0.92	0.78	0.67	0.71			0.67
Avg. % 'S'		0.036	0.018	0.031	0.031	0.023	0.027	0.027	
Min/Max		0.010/0.082	0.010/0.023	0.020/0.056	0.020/0.057	0.010/0.040			
Avg. % MgO		7.46	7.84	7.30	6.94	7.87	7.48		
Min/Max		6.74/8.31	7.77/7.91	6.99/7.73	6.32/7.45	7.57/8.58			
Avg. % Al2O3		16.47	16.58	17.52	17.79	17.71	17.21		
Avg. % FeO		0.87	0.77	0.87	0.96	0.76	0.85		
Avg. % K ₂ O		0.42	0.39	0.49	0.31	0.42	0.41		
BASICITY(-)		1	1.06	1.01	1.04	0.97	1.02		
Min/Max		0.94/1.02	1.06/1.06	0.98/1.03	0.95/1.08	0.9/1.05			
Avg. % 'P'		0.14	0.18	0.14	0.14	0.12	0.14		
Min/Max		0.13/0.16	0.16/0.18	0.13/0.18	0.12/0.17	0.12/0.13			

Fce.No.	Si→1.00	Si→0.85	No.of Casts	Tap to Tap >1:00 Hrs
4	1	5	8	1
5	1	2	4	1
6	1	5	6	0
7	1	3	10	0
SHOP	4	15	28	2

H ₂ IN TOP GAS						
	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8
G		3.70	STOP	4.50	3.10	
CO/CO ₂ RATIO						
G		1.25		1.34	1.47	
% Gas Util.						
G		44.39		42.79	40.5	45.5

MIXER	%Si	%Mn	%Sul	%P
Mixer-III				
Mixer-IV				

PELLET CHEMICAL AND SEIVE ANALYSIS

	Fe	SiO ₂	Al ₂ O ₃	CaO	MgO	+ 10 MM	- 5 MM	M.Size	Basicity	Slag Comp.
BF # 8 =>	61.23	6.42	4.62	0.68	0.30	52.80	1.90	11.00	0.11	12.02
BF # 7 =>	61.92	5.85	3.52	1.46	0.18				0.25	11.01
BF # 5 =>									#DIV/0!	0.00
BF # 4 =>									#DIV/0!	0.00

SKIP IRON ORE				
Fe (%)	SiO ₂ (%)	Al ₂ O ₃ (%)	Moist.	
BF # 4	60.88	4.58	4.78	4.4
BF # 5	61.00	5.57	3.70	
BF # 6	61.83	4.60	3.58	3.6
BF # 7	61.58	5.35	2.79	4.2
BF # 8	63.56	2.09	4.03	4
Avg.	61.77	4.44	3.78	4.05

BF SKIP I/ORE SEIVE ANALYSIS			
Fce.No.	+40 mm	- 10 mm	M.Size
BF # 4	19.00	11.00	26.85
BF # 5	24.70	13.10	27.90
BF # 6	19.05	10.35	26.35
BF # 7	17.77	9.77	27.60
BF # 8	21.02	13.75	26.87
AVG.	20.31	11.59	27.11

% FINES IN BF SKIP SINTER				TOTAL Fe			% FINES IN BF SKIP COKE			
Fce.No.	-10 mm	- 5 mm	M.Size	SHIFT A	SHIFT B	SHIFT C	Fce.No.	-40 mm	- 25 mm	M.Size
BF # 4	32.75	7.00	14.50	49.46	50.66		BF # 4	31.25	1.25	52.75
BF # 5	31.50	4.90	15.10			51.64	BF # 5	32.90	2.00	51.20
BF # 6	32.15	5.10	16.15	51.16	50.98		BF # 6	32.20	1.70	52.30
BF # 7	30.30	6.13	15.47	51.37	50.28	51.14	BF # 7	31.20	1.60	51.47
BF # 8	30.03	3.67	15.78	49.80	50.83	51.02	BF # 8	41.83	2.08	43.17
AVG.	31.35	5.36	15.40				AVG.	33.88	1.73	50.18

COKE QUALITY									
PROXIMATE ANALYSIS					COLD STRENGTH			HOT STRENGTH	
	Moisture	Ash	V.M	F.C	M-40	M-10	Sulphur	CSR	CRI
CSP-I: A	3.8	15.2	1.0	83.4			0.62		
CSP-II: A	4.1	14.6	0.9	84.0	80.4	7.2	0.55	64.1	23.8
CSP-III: A	5.0	14.9	0.9	83.7			0.60		
CSP-IV: A	0.4	15.2	1.0	83.4	87.4	5.8	0.57		
BF # 8 STOCK HOUSE SURFACE MIX.COKE ==>								65.6	23.5

COAL BLEND (%)													
	DATE	PCC	MCC	Aus(H)	BENGA	LVHR (H)	TUHUUP(hard)	Russian (S)	AUS(S)	USA(H)	USA(S)	% IMP	ASH %
CHP - I	3-Dec-24	11.0	3.0	36.0	7.0	9.0	0.0	11.0	10.0	13.0	0.0	86.0	CHP - I 11.2
CPP - II	3-Dec-24	11.0	3.0	35.0	7.0	9.0	0.0	0.0	11.0	13.0	11.0	86.0	CPP - II 10.6

CHEMICAL ANALYSIS

SINTER PLANT-2											
NORM(Basicity)					6 Kg/T of Sinter						
NORM(Mgo)					30 Kg/T of Sinter						
(1.90 - 2.00) (09-12-24) Nut Coke											
(2.75 - 3.25) (05-07-24) MILL SCALE											
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component
AO	49.08	9.05	8.05	2.97	14.73	3.12	1.83	6.68	OUT	1.83	28.45
A1	49.72	9.05	8.15	2.31	14.99	3.28	1.84	6.84	OUT	1.84	C I (SP # II)
A2	50.40	9.05	7.94	2.34	14.66	3.18	1.85	6.72	OUT	1.85	Coke
BO	49.62	9.13	8.32	2.54	15.11	3.09	1.82	6.79	OUT	1.82	Flux
B1	49.35	9.13	8.29	2.18	15.39	3.19	1.86	7.10	OUT	1.86	DTI
B2	49.13	9.13	8.64	2.19	15.66	3.20	1.81	7.02	OUT	1.81	
CO	49.77	9.13	8.22	2.20	14.86	3.08	1.81	6.64	OUT	1.81	
C1	50.37	9.13	7.74	2.64	14.34	2.96	1.85	6.60	OUT	1.85	
C2	51.47	9.13	7.84	1.57	13.90	3.16	1.77	6.06	OUT	1.77	
DAY AVG.	49.88	9.10	8.13	2.33	14.85	3.14	1.83	6.72	OUT	1.83	

SINTER PLANT-3											
NORM(Basicity)					0 Kg/T of Sinter						
NORM(Mgo)					12 Kg/T of Sinter						
(1.90 - 2.00) (09-12-24) Nut Coke											
(2.75 - 3.25) (05-07-24) MILL SCALE											
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component
AO	51.70	9.20	7.47	1.99	13.13	2.75	1.76	5.66	OUT	1.76	26.97
A1	50.94	9.20	8.04	2.39	13.79	3.02	1.72	5.75	OUT	1.72	
A2	51.06	9.20	7.72	2.26	13.94	2.84	1.81	6.22	OUT	1.81	
BO	50.61	9.04	8.40	2.07	14.10	2.88	1.68	5.70	OUT	1.68	
B1	51.67	9.04	7.51	1.98	13.53	3.08	1.80	6.02	OUT	1.80	
B2	50.17	9.04	8.10	2.06	14.91	2.81	1.84	6.81	OUT	1.84	
CO	50.30	9.04	8.19	2.18	14.67	2.91	1.79	6.48	OUT	1.79	
C1	50.21	9.04	8.15	2.35	14.92	2.81	1.83	6.77	OUT	1.83	
C2	52.08	9.04	7.48	1.84	13.79	2.63	1.84	6.31	OUT	1.84	
DAY AVG.	50.97	9.09	7.90	2.12	14.09	2.86	1.79	6.19	OUT	1.78	50.59

SINTER PLANT-3												
NORM(Basicity)					0 Kg/T of Sinter							
NORM(Mgo)					12 Kg/T of Sinter							
Sample ID	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Comp	50.95
SP-3/5/MC/2A0	51.73	9.84	7.36	2.51	13.06	2.75	1.77	5.70	OUT	1.77	27.21	
SP-3/5/MC/2A1	51.21	9.84	7.99	2.11	13.56	2.99	1.70	5.57	OUT	1.70		
SP-3/5/MC/2A2	51.45	9.84	7.60	2.41	13.65	2.77	1.80	6.05	OUT	1.80	C I (SP # III)	
SP-3/5/MC-2/B0	50.38	9.58	8.51	2.40	13.97	3.02	1.64	5.46	OUT	1.64		
SP-3/5/MC-2/B1	50.45	9.58	7.86	2.43	14.39	3.16	1.83	6.53	OUT	1.83	Coke	Flux
SP-3/5/MC-2/B2	50.24	9.58	8.02	2.28	14.79	2.93	1.84	6.77	OUT	1.84		
SP-3/5/MC-2/C0	51.09	9.58	7.80	2.84	13.91	2.74	1.78	6.11	OUT	1.78		
SP-3/5/MC-2/C1	49.44	9.58	8.62	2.20	15.85	2.80	1.84	7.23	OUT	1.84		
SP-3/5/MC-2/C2	52.31	9.58	7.31	2.24	13.17	2.93	1.80	5.86	OUT	1.80		
DAY AVG.	50.92	9.67	7.90	2.38	14.04	2.90	1.78	6.14	OUT	1.78		

BLAST FURNACE DEPARTMENT

Stoppage, L.Blast, Reduced Blast & offrod Duration and Reasons for Dated

1-1-25

on Dtd

2-Jan-2025

8:48

F.N	Duration			REASONS	ST	LB	RB	FB	Down Time	Furnace No	1	4	5	6	7	8	TOTAL
I				== FCE. BLOWN OUT ==						ABP	0	77000	77000	79000	119000	247000	599000
IV	00:00	03:00	LT. SKIP PIT I/ORE SPILLAGE			3:00				OPP	0	77000	77000	79000	95900	221100	550000
			ORE GATE HOPPER COVERING PLATE BROKEN						M/Rate	0	69719	22816	89993	117273	248651	548452	
									Weekly Plan	0	7350	6400	6300	11700	24300	56050	
V	06:00 08:00 09:00 10:00 19:00 20:00 01:00	08:00 09:00 10:00 19:00 20:00 03:00	TO DRY THE FCE.			1:00	2:00	21:00:00	0:45:00	Prod.(Cumm.)	0	6610	4826	6277	11996	24752	54461
			==DO==			1:00			Fulfillment	#DIV/0!	89.9	75.4	99.6	102.5	101.9	97.2	
			SCHD. MAINT. FOR T-8 SHELL WELDING, T-5 CHANGING	9:00				Today'sPlan	0	2550	2550	2600	3900	8100	19700		
			T-2 COMPENSATOR REPAIR JOB					HM Prod	0	2249	736	2903	3783	8021	17692		
			T-17 COMPENSATOR CHANGED					Cu.Prod	0	2249	736	2903	3783	8021	17692		
			CUT OFF VALVE PATCH WORK, BLT INSPECTION JOB					D/Rate	0	2249	736	2903	3783	8021	17692		
			PR. GR. INCREASED			1:00			No.OF SKIPS (6:00 TO 6:00)						Charge	Total	
			==DO==						A		209	25	208	249	39	691	
			RT. SINTER CLOSING PROB. MOTOR CHANGED					4:00		B		226	65	235	232	46	758
			CROSS CHARGING (U/ELECT.)					2:00		C		138	178	221	243	46	780
										Total skips	0	573	268	664	724	131	2229
							9:00	2:00	9:00	4:00:00		Stoppage on Date			9:00		
VI	07:30 02:00	09:30 03:00	RT. SINTER FINES FEEDER MOTOR F BOLT							Low Blast on Dt			2:00				2:00:00
			RT. ORE BELT SWAY (U/MECH.)			2:00			Reduced blast on Dt		3:00	9:00	3:00		1:00:00	16:00:00	
			CHARGING DELAY, LT. ORE GATE JAM			1:00			No of Tappings	0	8	4	6	10	12	40	
						3:00	21:00:00	0:45:00	No of Not Dry casts	0	0	0	0		0	0	
VII			UPTAKE ON LOWER SIDE						Coke Rate On Date	497	609	470	457	385	438		
							24:00:00		Coke Rt for month	497	609	470	457	385	438		
VIII	08:00	09:00	MULTIPLE CHARGING PROB. MCB & MGB PROB. (ELECT.)			1:00				Hot Blast Temp.	984	910	1000	1008.125	1142	889	
			LMG PROB. (ELECT.)						Metal ladle position			Slag ladle position					
						1:00	23:00:00	0:15:00	In circulation	40	+80	11	Total	In Use	Under M/Maint.	Under E/Maint.	
			TOTAL	9:00:00	2:00:00	16:00:00	93:00:00	14:00:00	Open Hearth	22	+75	4	32	32			
									Reserve		+70	3					

COAL DUST INJ. (Kg/THM)				
BF				
No.	DAILY/ Dat	CDI Rate(kg/thm	HM Production on that date(T)	CDI in tonnage(T)
1	25.03.2021	124	1852	230.15
4	14.12.2022	226	1250	281.90
5	03.11.2024	209	1118	233.62
6	10.10.2022	206	1821	313.14
7	07.08.2024	158	3083	488.15
8	05.11.22	192	7651	1472.00
SHOP	21.10.24	160	14750	2360.00

CDI On Tonnage Basis			
BF			
No.	DAILY	Best CDI On tonnage Basis(T)	HM Production on that date(T)
1	25.03.2021	230.15	1852
4	18.10.2024	410.99	1954
5	07.11.2024	427.53	2304
6	19.09.2022	334.99	2044
7	10.12.2022	540.30	4176
8	30.11.2022	1575.97	8793
SHOP	25.01.2023	2721.79	18771

Annual ABP Target(FY 2024-25)			Till Date
HM Target(MT)	6.60	4.38	
Coke RATE(kg/thm)	429	432	
CDI Rate(kg/thm)	133	139	
FUEL RATE(kg/thm)	588	593	
Financial Data			
Cost of HM Production in FY 2023-24 (per ton of HM)	Rs 35524		
Cost of HM Production in FY 24-25 till Oct'24(per ton of HM)	Rs 33554		
Cost of HM Production in Oct'24(per ton of HM)	Rs 32291		
Contribution Margin per ton (Profit) for Pig Iron in oct-24	Rs 8995		
Cost of Skip coke in FY 24-25 Till Oct'24 (per ton)	Rs 38175		
Cost of Skip Coke in Oct'24(per ton)	Rs 36476		
Previous Year ABP vs Actual (FY2023-24)	ABP	ACTUAL	
HOT METAL(MT)	6.20	5.99	
COKE RATE(kg/thm)	430	455	
CDI RATE(kg/thm)	125	120	
Fuel Rate(kg/thm)	587	597	
Pig Iron Produced in FY 2023-24	67695 T		
Pig Iron Produced in FY 2024-25	26673 T		

RECORDS						
B.Fces.	Hot Metal Production		COKE RATE		CDI RATE	
	MONTHLY		MONTHLY		MONTHLY	
BF # 1	61574	AUG.'17	436	MAR.'12	124	JAN.'24
BF # 4	82493	MAY.'24	438	MAR.'23	154	OCT.'24
BF # 5	82758	MAR.'22	418	FEB.'09	162	OCT.'24
BF # 6	89535	JAN.'09	397	JAN.'02	144	OCT.'22
BF # 7	128247	OCT.'17	428	MAR.'19	124	NOV.'22
BF # 8	252731	DEC.'24	370	SEP.'21	160	DEC.'23
SHOP	555571	DEC.'24	414	SEP.'21	150	OCT.'24
PCM POURING	41	2451	30-06-2023			